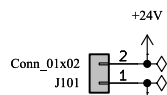
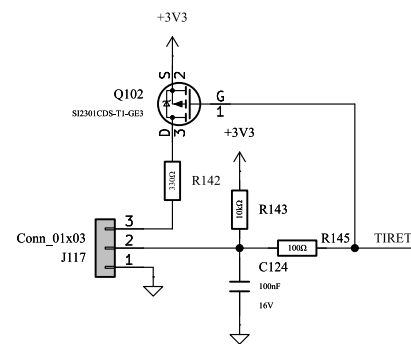
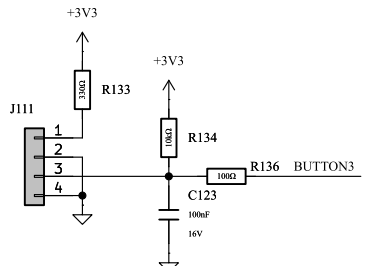
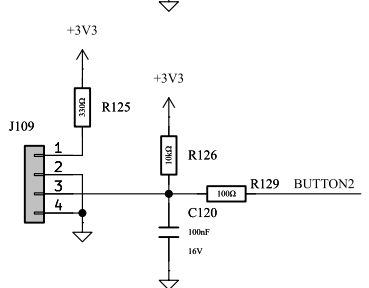
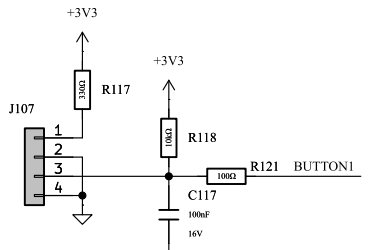
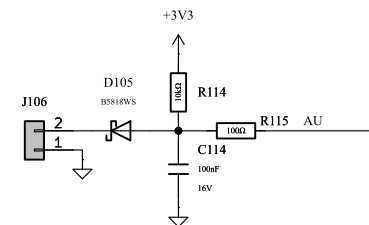


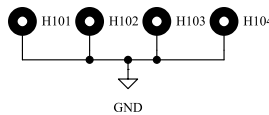
24V Input



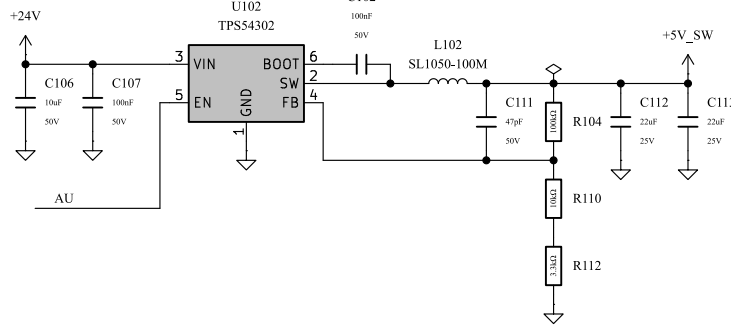
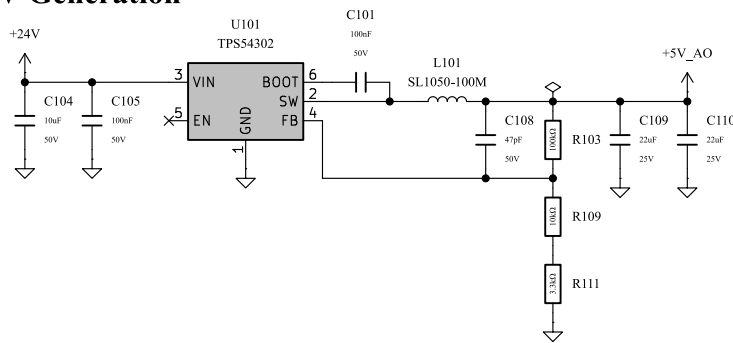
Discrete Inputs



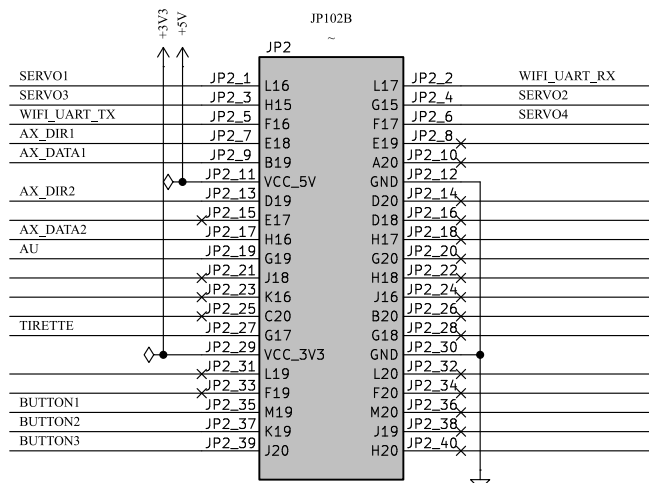
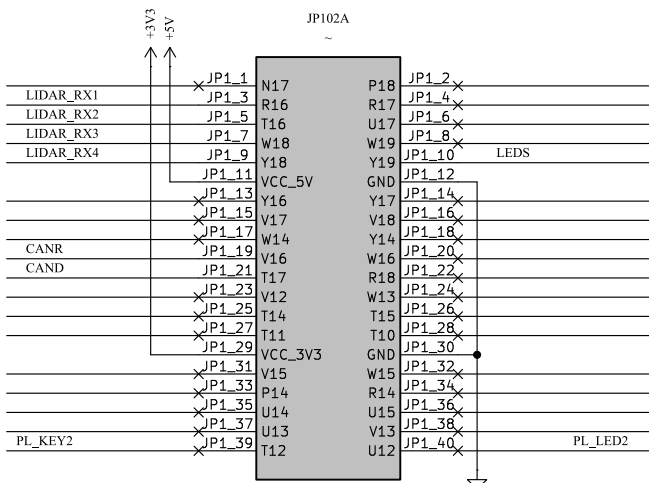
Mounting holes



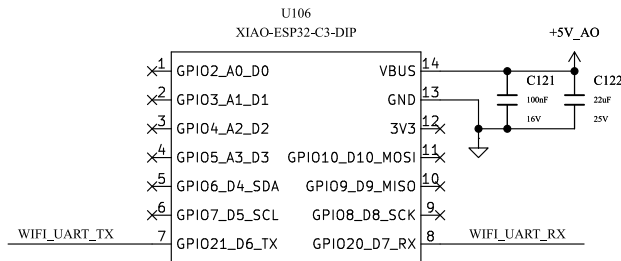
5V Generation



Zynq Z7 Lite Connector



Wi-Fi Comm



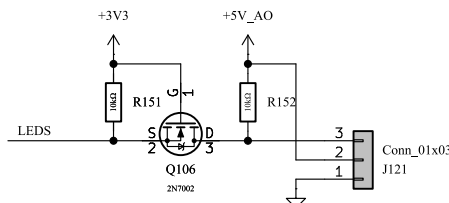
24V generates two 5V
- AO (Always On, for LED, LIDARs and Wi-Fi) - NO AU
- SW (Switched, for servoes) - AU

Supplied by FPGA:
- 3V3 Logic and Interfaces (Low Power)
- 5V Logic and Interfaces (Low Power)

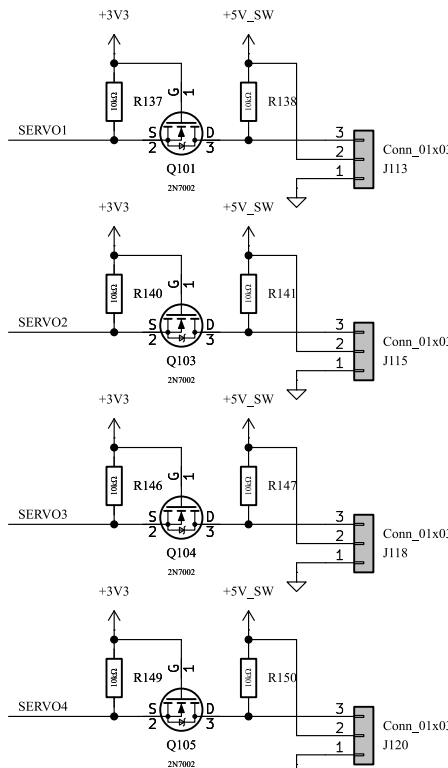
I/Os:
3V3:
- IN: Buttons (x3)
- IN: AU (Emergency Stop)
- IN: Tirette
- INOUT: CAN
- INOUT: Wi-Fi

5V:
- IN: LIDARs
- INOUT: AX12 (x2)
- OUT: LEDs
- OUT: Servos

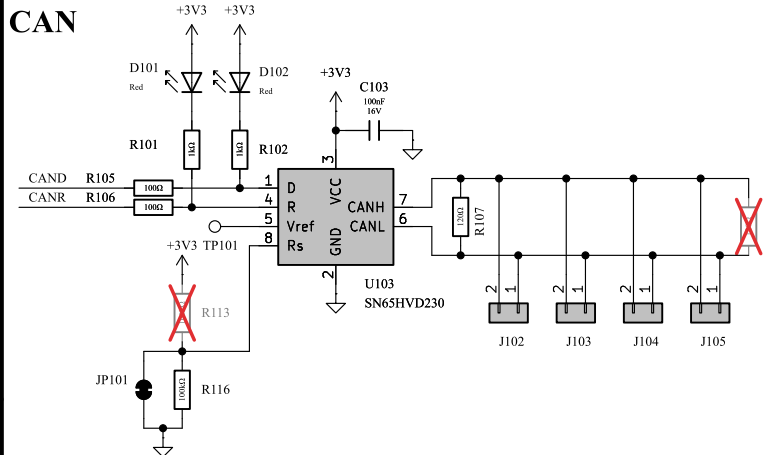
5V LED Strip



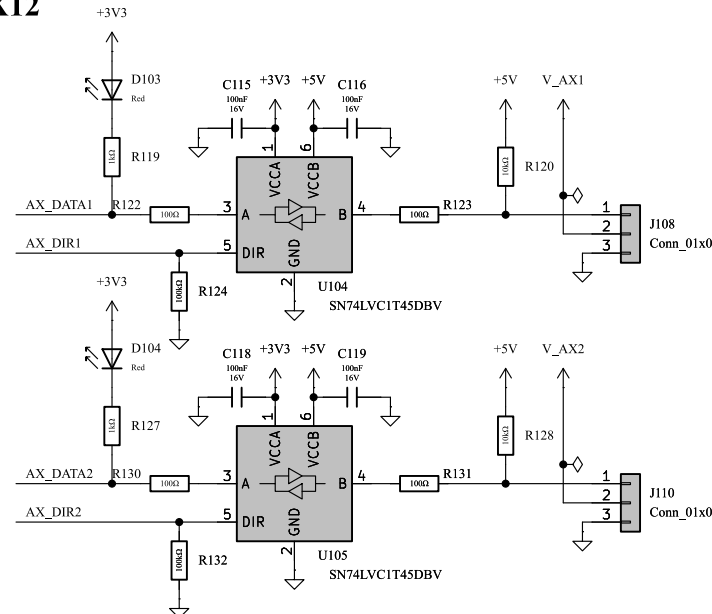
5V Servos



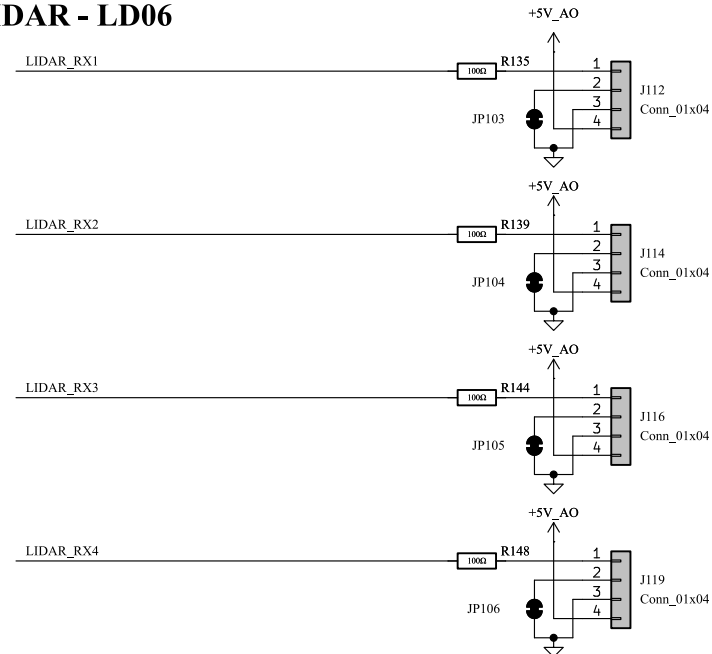
CAN



AX12



LIDAR - LD06



Les Opossums – Gabriel de Melo Souza

Sheet: /
File: ELE_MB_A1.kicad_sch

Title: Motherboard

Size: A3 Date: 2026-02-16
KiCad E.D.A. 9.0.7

Rev: A1
Id: 1/1